

Machine Learning & Deep Learning Project Report

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1. Introduction

The purpose of this project is to explore how machine learning and deep learning models can be used to **predict heart disease based on patient medical data**. Early detection of heart disease is **very important in healthcare** because it helps identify high-risk patients **before serious complications occur**.

2. Dataset Description

The dataset contains 920 patient records with 16 medical features such as age, sex, chest pain type, cholesterol, and blood pressure.

The dataset contains missing values in several columns such as ca, thal, slope, and chol, which required preprocessing before training.

The target variable is binary:

- **1 → Heart disease present**
- **0 → No heart disease**

The dataset is approximately balanced with:

- **Positive cases: 509**
- **Negative cases: 411**

Dataset shape: 920 rows × 16 features (before preprocessing)

After preprocessing: X features increased due to one-hot encoding

3. Data Preprocessing

Missing values were handled using median for numerical features and mode for categorical features

Categorical variables were converted into numerical values using one-hot encoding

Finally, the dataset was split into training (80%) and testing (20%) sets

Feature scaling was applied using StandardScaler for both machine learning models and neural network to ensure fair comparison

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4. Machine Learning Models

Three machine learning models were used

Logistic Regression

Random Forest

K-Nearest Neighbors (KNN)

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Each model was evaluated using Accuracy, Precision, Recall, and F1-score

KNN achieved the best performance among traditional machine learning models based

on F1-score, making it the most reliable classical model for this dataset

5. Neural Network Model

A neural network was built using Keras with two hidden layers and Dropout to reduce overfitting.

The model was trained for 10 epochs using the Adam optimizer.

The neural network performed competitively but did not outperform KNN.

This is expected because tabular datasets often perform better with traditional machine learning models.

6. Results & Comparison

KNN achieved the highest F1-score among all models, while the neural neural network performed slightly lower in terms of F1-score. This shows that for structured medical data, simpler models can sometimes outperform deep learning models

7. What I Learned

This project helped me **understand the full machine learning pipeline from raw data to final prediction**

I learned the importance of data preprocessing and **how it affects model performance**

I also learned that complex models are not always necessary, **and simpler models can still perform well**

I learned that model performance depends more on data type than model complexity

Conclusion

In conclusion, the project successfully built and compared multiple machine learning models and a neural network

KNN performed best among all models, while the neural network also achieved strong results